Gravity feed-forward vs. position tracking on the MABEL left arm

Hardware-bridge experiment `gravity_ff_tracking`

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Question

Does gravity feed-forward measurably improve joint position tracking on the real arm, and by how much? The concern is the classic failure of a PD servo: it *sags* under a sustained load and cannot be tuned out.

Why tuning cannot fix it

The arms run the DaMiao MIT law $T = k_p(q^* - q) + k_d(\dot{q}^* - \dot{q}) + t_{\mathrm{ff}}$. At a static hold $\dot{q} = 0$, so the motor must supply the joint's gravity torque $\tau_g(q)$:

$$k_p\,(q^* - q) + t_{\mathrm{ff}} = \tau_g(q) \quad \Longrightarrow \quad e_{\mathrm{ss}} = \frac{\tau_g(q) - t_{\mathrm{ff}}}{k_p}.$$

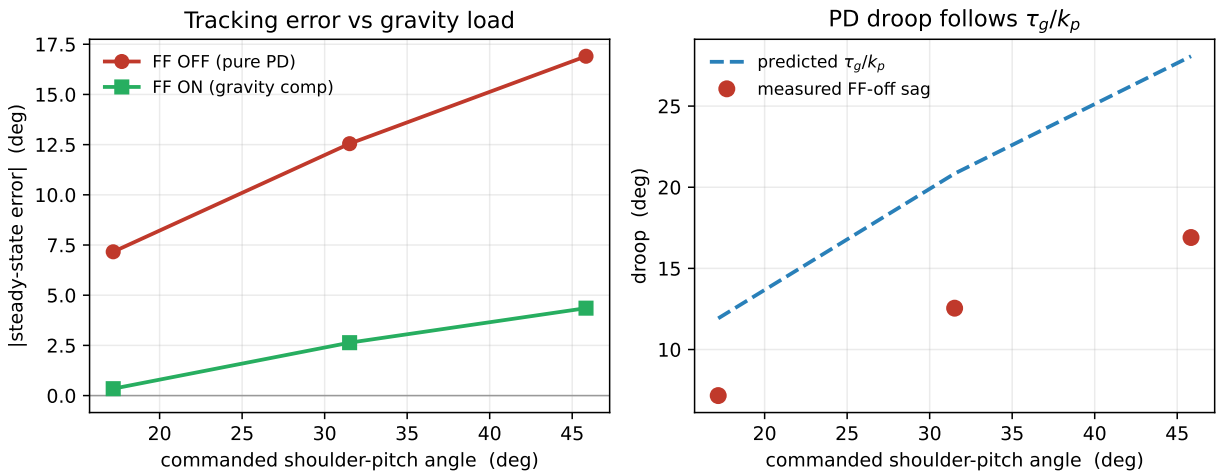
With no feed-forward ($t_{\mathrm{ff}} = 0$) the steady-state error is a pure droop $e_{\mathrm{ss}} = \tau_g(q)/k_p$. It is **not** a tuning problem: DaMiao packs k_d into 12 bits over $[0, 5]$, so $k_d \le 5$ is a hardware cap; a critically damped loop then caps $k_p \approx 15$ on the shoulder. A bigger gain is simply unavailable. A *correct* feed-forward $t_{\mathrm{ff}} = \tau_g(q)$ cancels the load and drives $e_{\mathrm{ss}} \to 0$ regardless of k_p.

Method

The left shoulder-pitch joint (`left_arm_2`, $k_p = 14.9$) was commanded to three elevated angles where gravity actually loads it (τ_g from 3.1 N m hanging to 7.3 N m lifting out). Each target was held 12 s; the steady-state angle is the median of the last 3 s. Targets were sent over the `:9090` `joint_command` wire message — the *same* path the iOS app uses — so this also confirms the operator control path end to end. The run was executed twice, once with `MABEL_HW_ARMS_FF_SCALE=1` and once `=0`, and the FF state of each pass was verified against the running process (a first automated attempt left FF on in both passes; that run was discarded).

Result

target	FF OFF (pure PD)		FF ON (grav. comp)		improvement
(deg)	settled (deg)	error	settled (deg)	error	
17.2	10.0	−7.2°	16.8	−0.3°	6.8°
31.5	19.0	−12.5°	34.1	+2.6°	9.9°
45.8	28.9	−16.9°	50.2	+4.4°	12.5°



Conclusion

Gravity feed-forward reduces the steady-state tracking error from as much as **17**° of droop to within **4**°, and the FF-off droop tracks the predicted τ_g/k_p — confirming the sag is the gravity term, not friction or a control bug. The remaining FF-on error is a slight *over*-shoot (+4.4°) at the highest angle, i.e. the model lifts marginally too hard there; this is the one open knob (the shoulder-mount tilt, 0° on the metal vs. −10° in the MJCF, and the hands-removed mass model). It is small and can be trimmed with `MABEL_HW_ARMS_FF_SCALE` slightly below 1 or by revisiting the mount angle. **Net: gravity compensation is working and is essential — without it the arm cannot hold a lifted pose.**